

Karanvir Singh Prihar

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SUMMARY

Energy Data Analyst with a mechanical/aerospace engineering foundation (M.Eng, University of Toronto) and three years of hands-on residential energy-efficiency consulting experience, including 250+ HOT2000-modelled home energy audits delivered under Canada's Greener Homes program. Currently completing an intensive, self-directed upskilling program in SQL, Python, and Power BI, with an end-to-end analytics project examining Ontario's electricity generation mix already published. Brings a rare combination of energy-sector domain expertise and analytical rigor to turn field, program, and utility data into clear, decision-ready insights.

HIGHLIGHTS

- Rare combination of hands-on residential energy audit experience (250+ homes) and growing SQL, Python, and Power BI proficiency, positioned to bridge field energy data and analytics.
- Built and published an end-to-end data analytics project examining Ontario's electricity generation mix using SQL Server, Python, and interactive visualizations.
- Delivered detailed technical energy assessments and retrofit reports to 250+ homeowners under Canada's Greener Homes program, translating modelling outputs into clear recommendations.
- Applied structured, data-driven problem-solving (Global 8D, DMAIC) to lift production quality run rate from 85% to 96% in a prior engineering role.
- Engineering foundation (M.Eng Aerospace, University of Toronto) combined with intensive, self-directed upskilling in modern data tools.

TECHNICAL SKILLS

Energy & Building Science: HOT2000 Modelling | Blower Door Testing | Residential Retrofit Assessment | Canada Greener Homes Program

BI & Visualization: Power BI (DAX, Power Query) | Excel | Plotly

Languages & Databases: SQL | Python (pandas, seaborn) | SQL Server

Tools: Git | GitHub

PROJECTS

Ontario's Energy Mix: The Accelerating Reliance on Natural Gas

Jun 2026

github.com/ksprihar/ontario-energy-mix

- Built an end-to-end SQL and Python analysis of Ontario's electricity generation mix (2015–2026) using public IESO data, identifying that Natural Gas has shifted from a peaking fuel into a structural baseload fallback, absorbing +1,232 GWh of generation growth against a broader system increase of +1,138 GWh since 2024.
- Automated data ingestion from IESO's public XML/CSV report endpoints using Python (requests, pandas), and designed a SQL Server schema with integrity validation (null, duplicate, and continuity checks) to ensure analysis-ready data.
- Wrote T-SQL queries using window functions and CTEs (DENSE_RANK, rolling 12-month averages, window-of-aggregate patterns) to quantify Gas's rising share of generation during peak-demand periods — from 7.8% in low-demand months to 18.0% in the most extreme.

Ontario BPS Energy & GHG Dashboard

Jul 2026

github.com/ksprihar/ontario-bps-energy-dashboard

- Built an end-to-end Power BI dashboard (star schema, DAX, Power Query) on a messy, regulator-published Ontario government dataset covering ~18,000 properties across 3 years.

- Designed a 4-page report (plus 3 hidden tooltip pages) with dynamic Field Parameters, custom tooltips, and a two-page navigation architecture solving a structural Power BI drillthrough limitation.
- Diagnosed and resolved multiple non-obvious Power BI bugs (a Desktop-to-Service color-persistence issue, a RANKX ranking-tie edge case) and quantified a real ID-reconciliation data-quality problem rather than silently merging records.

PROFESSIONAL EXPERIENCE

Career Transition – Data Analytics Upskilling

Jan 2025 – Present

- Relocated internationally (Canada → India → Canada) while completing a self-directed, full-time upskilling program in SQL, Python, and Power BI to pivot from energy consulting into energy-sector data analytics.
- Completed Angela Yu's "100 Days of Code" Python course in full (certificate of completion) and built a daily SQL problem-solving practice on StrataScratch and DataLemur.
- Designed, built, and published an end-to-end data analytics portfolio project analyzing Ontario's electricity generation mix (SQL Server, Python, Docker, GitHub Pages).
- Currently completing a Power BI/DAX coursework and building a second Power BI project using Ontario's Broader Public Sector Energy & GHG Reporting dataset, while preparing for the PL-300 exam (Aug 2026).

Registered Energy Advisor – Windfall Ecology Centre

Apr 2022 – Jun 2025

- Conducted 250+ comprehensive residential energy audits – including blower-door testing, HOT2000 energy modelling, insulation and HVAC assessments, and building envelope analysis – for homeowners across Ontario under Canada's Greener Homes program.
- Analyzed and interpreted building performance data to identify efficiency upgrade opportunities, translating technical modelling outputs into clear, actionable retrofit recommendations for 250+ households.
- Delivered detailed audit reports and cost-benefit breakdowns, advising homeowners on retrofit prioritization and available incentive programs.
- Maintained up-to-date technical knowledge of HER+ program requirements and evolving construction/retrofit methods through ongoing training.

Operations and Production Manager – Tata Motors Limited

Aug 2016 – Oct 2018

- Improved Quality Direct Run Rate (DRR) from 85% to 96% by applying structured, data-driven problem-solving methods (Global 8D, DMAIC) to identify and eliminate recurring defect trends.
- Reduced production line cycle time by 25% (160s → 120s) over one year through continuous, metrics-driven process improvement (Kaizen).
- Directed daily operations and a team of 100+ employees, delivering 30,000+ vehicles/year (~\$250M in annual sales) while coordinating cross-functionally with supply chain teams.
- Reduced shop-floor safety incidents by 50% through systematic tracking and elimination of hazards.

EDUCATION

Master of Engineering, Aerospace Engineering – University of Toronto

2018 – 2020

Bachelor of Technology, Mechanical Engineering – IIT Ropar

2012 – 2016

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300) – In Progress, exam targeted August 2026

- 100 Days of Code: The Complete Python Pro Bootcamp – Certificate of Completion